

```

import matplotlib as mpl
import matplotlib.font_manager as fm
import os

# — Registrar Arial (detecta automaticamente no sistema) —————
def _registrar_arial():
    fontes = [f.name for f in fm.fontManager.ttflist]
    if "Arial" in fontes:
        return

    candidatos = [
        r"C:\Windows\Fonts\arial.ttf",
        "/Library/Fonts/Arial.ttf",
        "/usr/share/fonts/truetype/msttcorefonts/Arial.ttf",
        os.path.join(os.path.dirname(__file__), "Arial.ttf"),
    ]
    for caminho in candidatos:
        if os.path.exists(caminho):
            fm.fontManager.addfont(caminho)
            return

    import urllib.request
    dest = os.path.join(os.path.expanduser("~"), "Arial.ttf")
    if not os.path.exists(dest):
        url = "https://github.com/matomo-org/travis-scripts/raw/master/fonts/Arial.ttf"
        urllib.request.urlretrieve(url, dest)
        fm.fontManager.addfont(dest)

_registrar_arial()
mpl.rcParams["font.family"] = "Arial"

import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.colors import LinearSegmentedColormap

# — Estilo —————
sns.set_style("white")
sns.set_context("talk")
mpl.rcParams["font.family"] = "Arial" # reafirmar após set_context (seaborn sobrescreve)

# — Dados —————
tabela = df1["Óbito"].value_counts().to_frame().T

# — Mapa de cores —————
cores = LinearSegmentedColormap.from_list(
    "verde_laranja",
    ["#FFCB99", "#ffe29a", "#A8D8C8"]
)

# — Figura —————
plt.figure(figsize=(10, 8))

ax = sns.heatmap(
    tabela,
    annot=True,
    fmt="d",

```

```

cmap=cores,
linewidths=2,
linecolor="white",
annot_kws={
    "size": 18,
    "weight": "bold",
    "color": "black",
    "fontfamily": "Arial",    # anotações dentro das células
},
cbar_kws={
    "label": "Number of patients",
    "shrink": 0.8,
}
)

```

— Colorbar

```

cbar = ax.collections[0].colorbar
cbar.ax.yaxis.label.set_rotation(270)
cbar.ax.yaxis.set_label_coords(5.5, 0.5)
cbar.ax.yaxis.label.set_fontfamily("Arial")    # rótulo da colorbar
cbar.ax.yaxis.label.set_fontsize(18)
for lbl in cbar.ax.get_yticklabels():          # tick labels da colorbar
    lbl.set_fontfamily("Arial")

```

— Fundo

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ax.set_facecolor("#f5f5f5")

```

— Título

```

plt.title(
    "Distribution of Outcome",
    fontsize=25,
    weight="bold",
    pad=20,
    fontfamily="Arial",
)

```

— Eixos

```

plt.xlabel("Death or Discharge", fontsize=16, weight="bold", fontfamily="Arial")
plt.ylabel("", fontfamily="Arial")

plt.xticks(fontsize=13, weight="bold")
plt.yticks(rotation=0)

for lbl in ax.get_xticklabels():
    lbl.set_fontfamily("Arial")
for lbl in ax.get_yticklabels():
    lbl.set_fontfamily("Arial")

```

— Layout & Salvar

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plt.tight_layout()
plt.savefig("heatmap_obito_alta.png", dpi=180, bbox_inches="tight")
plt.show()

```